# Doc4TF
# Automatic Generation of Text-Fabric Documentation

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**At a glance**

- **Purpose:** automatically generate a hyperlinked documentation set for a Text-Fabric (TF) dataset.
- **Input:** an existing TF dataset (nodes, edges, and features) plus its embedded feature metadata.
- **Output:** per-feature pages and index pages; optionally rendered as HTML for direct browsing.
- **Typical use:** bootstrap missing documentation and validate that existing documentation matches the data.

## 1 About Doc4TF

Doc4TF (`https://github.com/tonyjurg/Doc4TF`) is a Jupyter Notebook that contains Python code to automatically generate a documentation set for any Text-Fabric dataset, based directly on the dataset contents. It can serve as a starting point for new documentation, as a substitute when documentation is missing, and as a tool for validating existing documentation. This software is provided "as-is" under a Creative Commons license.

Ideally, a comprehensive documentation set is developed alongside the creation of a Text-Fabric dataset. In practice, however, this is often left incomplete during the initial phase or not consistently updated after features change.

Text-Fabric represents textual data as a graph-based dataset, where words, phrases, clauses, and larger textual units are modeled as nodes connected by edges that capture their relationships. Unlike a flat table or sequential corpus, this abstract network of nodes and features is not immediately transparent when viewed in raw form. Clear documentation is therefore essential for understanding the structure and making the dataset accessible. Doc4TF addresses this need by extracting information from the dataset itself and presenting it across a set of interlinked pages. It focuses on two main kinds of information:

- Structural details about nodes (e.g., their names) and features (e.g., name, data type, feature type, and value frequencies).
- Descriptive details about features. The extent and correctness of these descriptions depend on the embedded metadata.

The pages produced by Doc4TF can serve as a robust starting point for developing a new documentation set, or as validation for an existing one. A major advantage is that the output is fully hyperlinked, which can be laborious to produce manually. In practice, Doc4TF produces:

- **Feature pages:** one page per feature, including structural details and (when available) a description taken from the embedded metadata.
- **Index pages:** overview pages that link to the individual feature pages, grouped by feature name, node type (e.g. word, sentence), data type (integer, string), and feature type (node or edge features).

Figure 1 shows the resulting structure as it is created by the Doc4TF script. The hyperlinks between the overview and the feature description pages are represented as arrows.

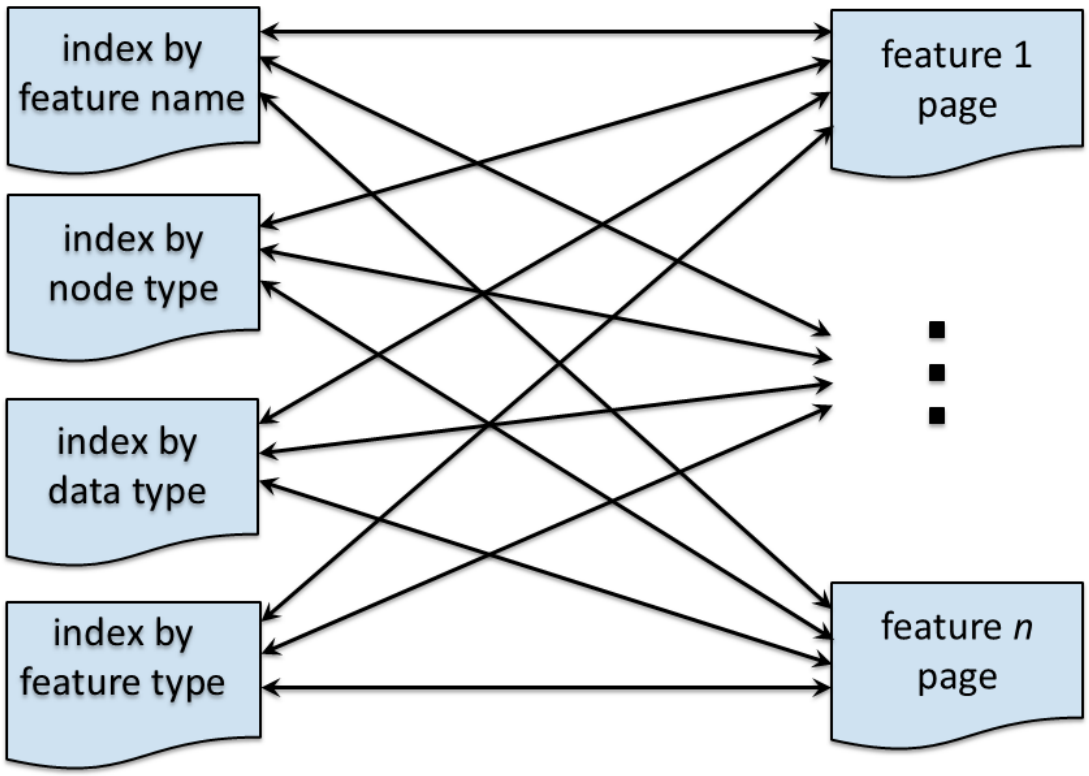


Figure 1: The relation between the pages created by Doc4TF

Figure 2 shows an example of a generated overview page. For any generated page, the information that is part of the `description` field is taken from the metadata that was provided during the creation of the dataset.

Doc4TF pages for Nestle 1904 Greek New Testament (version 0.5.9)

## Overview features by node type

Overview by name, data type, or feature type.

### book

| Feature | Feature type | Data type | Description | Examples |
|---|---|---|---|---|
| book | Node | String | book name (full name) | Acts Colossians Ephesians Galatians |
| bookshort | Node | String | book name (abbreviated) from ref attribute in xml | 1CO 1JN 1PE 1TH |
| lang | Node | String | language the text is in | el |
| num | Node | Integer | generated number (not in xml): book: (Matthew=1, Mark=2, ..., Revelation=27); sentence: numbered per chapter; word: numbered per verse. | 1 2 3 4 |

### chapter

| Feature | Feature type | Data type | Description | Examples |
|---|---|---|---|---|
| book | Node | String | book name (full name) | Acts Colossians Ephesians Galatians |
| chapter | Node | Integer | chapter number, from ref attribute in xml | 1 2 3 4 |

Figure 2: Example of a generated overview page

## 2 Requirements

- A Python environment capable of running Jupyter notebooks.
- The `text-fabric` Python package installed.
- A Text-Fabric dataset to document (with meaningful embedded metadata for best results).

## 3 Using Doc4TF

Since Doc4TF is implemented as a Jupyter notebook, you will need an environment capable of running Jupyter notebooks. As this tool is intended to be used alongside Text-Fabric, it is likely that you have already set up such an environment. If you have not done so yet, a good option is to install Anaconda.[1] Text-Fabric requires at least Python version 3.7.0,[2] so any Anaconda Navigator from version 5.3.0 (Sept 28, 2018) upwards should suffice.

Note that your environment should (for obvious reasons) include the Python package Text-Fabric. If it is not installed yet, it can be installed using `pip`. More details on installing the Text-Fabric package can be found in `tf.about.install`.[3]

[1] `https://www.anaconda.com/`.
[2] `https://annotation.github.io/text-fabric/tf/about/install.html`.
[3] `https://annotation.github.io/text-fabric/tf/about/install.html`.

To start using Doc4TF, first download the Jupyter notebook from the GitHub repository,[4] place it anywhere on your system where you can execute it, and run it. The notebook will guide you through the process, which typically consists of the following steps:

1. Load the Text-Fabric dataset you want to document.
2. Run the subsequent cells to extract nodes, features, and metadata into an internal Python data structure.
3. Generate one documentation page per feature and a set of index pages.
4. Pages are rendered as either markdown for storage on GitHub, HTML for direct browsing, or in both formats.

Markdown is the standard format for Text-Fabric feature documentation, but viewing it in a web browser requires post-processing. A common approach is to upload the files to a GitHub repository, where they are rendered automatically, or to install a browser extension such as markdown viewer for local use.[5]

Alternatively, the Doc4TF script can directly produce HTML files, which can be stored on a local drive and opened in any web browser without further pre-processing or additional extensions.

## 4 Legal

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## 5 Attribution

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For example:

- Jurg, T. (2024). Doc4TF. Zenodo. `https://doi.org/10.5281/zenodo.12705876`
- Jurg, T. (2024). Doc4TF. Available on GitHub: `https://github.com/tonyjurg/Doc4TF`

[4] `https://github.com/tonyjurg/Doc4TF/blob/main/CreateFeatureDoc.ipynb`.
[5] `https://github.com/simov/markdown-viewer`.
[6] `http://creativecommons.org/licenses/by/4.0/`.